

On February 13 & 14 2026, members of NECTA and ZEC in Tanzania attended a workshop regarding the Self-Administered EGRMA in Kiswahili and its use in evaluating the MsingiTek program. As part of this workshop, attendees collaborated to make decisions on removal of items from the assessment, specifically from the Spelling and Vocabulary tasks, to shorten the test without compromising reliability and validity. For the full original analysis including these items, see the Pilot Report: *Digital EGRA/EGMA Work Order No 4. Deliverable 4: Results of Pilot Test in Kiswahili* (RTI International, 2025).(<https://bit.ly/4cSjlie>).

Item Removal:

Spelling:

Removing Items 5, 6, 7, 11

Participants decided on removal of these 4 items, reducing the total Spelling items to 8

Item	Sentence
5 twende	Neno ni twende . Mwalimu amesema twende darasani. Twende. Twende.
6 wangapi	Neno ni wangapi . Darasa lenu lina wanafunzi wangapi? Wangapi. Wangapi.
7 shughuli	Neno ni shughuli . Tunawasaidia wazazi shughuli za nyumbani. Shughuli. Shughuli.
11 machungwa	Neno ni machungwa . Watoto walienda kuchuma machungwa . Machungwa. Machungwa.

Vocabulary:

Removing Items 2, 4, 12, 17, 20

Participants decided on removing these 5 items, leaving 15 total Spelling items

2	likoyo	kurunzi	ngakomi	kosoni
4	fyekeo	ganu	tala	rupe
12	kamula	kadri	pwiliko	twarama
17	theluji	tongoka	ruda	hekeresha
20	nyemung'u	selugi	nganye	tenzi

The following analysis shows the updated internal consistency, test-retest reliability, and assessment validity analyses after removal of the items. As shown below, the items removed had very little impact on the psychometric properties of the assessment.

Updated Outcomes:

The updated Outcomes from the literacy assessment are presented here. The average percent score for the Vocabulary subtask has increased slightly, and the average percent score for the Spelling subtask has only decreased by 0.2 percentage points. The original outcomes can be found in Exhibit 1 on page 9 of the Pilot report.

Updated SA-EGRA Average Percent Scores, by Task

SA-EGRA Task	Average Percent Score
Syntax	64.4%
Letter Sounds	83.5%
Vocabulary	71.3%
Spelling	68.7%
Silent Reading Comprehension	66.9%
Short Story Reading	67.0%
Comprehension	
Syllables	80.2%

Updated Internal Consistency:

Presented below are the updated internal consistency analyses for the overall Self-Administered literacy assessment as well as for the Vocabulary and Spelling tasks. In the overall Confirmatory Factor Analysis, factor loadings have not changed in any meaningful way after updating the two tasks. The original analysis can be found in Exhibit 4 on page 11 of the report. All subtasks continue to load highly on the first factor and show loadings exceeding the acceptability threshold of 0.3. The Letter Sounds task is still showing slightly lower loadings, which was noted in the report, and is most likely due to being a task geared towards beginners.

The Confirmatory Factor Analysis for the Spelling task shows the factor loadings for the final eight items. Again, there was very little change in the factor loadings for each item from the original analysis (found in Exhibit C8 in Annex C of the report). All items loaded highly and the task shows excellent internal consistency.

Internal consistency of the Vocabulary task was assessed using both Confirmatory Factor analysis and IRT item-level analysis. The original analysis can be found in Exhibit C3 in Annex C of the Report. In the updated analysis with the reduced item set, all items loaded highly on the first factor, with all items showing factor loadings over 0.3. All items also showed appropriate Discrimination, with scores well over 0.2, and an acceptable range of Difficulty scores between 0.61 and 0.80. The bi-serial correlation for the updated Vocabulary task showed that all items had appropriately strong and positive correlations to the scores on the overall task. Altogether, these results show that this task is still a suitably strong task that can be used to assess literacy skills in students.

Updated Factor Loadings for SA-EGRA Tasks

SA-EGRA Task	Factor 1 Loadings
Letter Sounds	0.4913
Syllables	0.6745
Short Story Reading Comprehension	0.7609
Silent Reading Comprehension	0.7942
Vocabulary	0.8453
Syntax	0.8207
Spelling	0.8033

**Updated Item Factor Analysis
for the *Spelling* task**

Item	Factor Analysis
1	0.6835
2	0.7370
3	0.8630
4	0.7575
8	0.7707
9	0.8382
10	0.8555
12	0.8073

**Updated Item Factor Analysis and
IRT parameters for the *Vocabulary* task**

Item Number	Factor Analysis	Item Response Theory		
		Discrimination	Difficulty	Bi-serial Correlation
1	0.7282	0.79	0.71	0.74
3	0.7767	0.76	0.73	0.78
5	0.4371	0.59	0.69	0.51
6	0.6037	0.71	0.71	0.63
7	0.6814	0.73	0.73	0.7
8	0.5017	0.67	0.67	0.56
9	0.6995	0.79	0.69	0.72
10	0.6807	0.75	0.7	0.7
11	0.5693	0.68	0.71	0.6
13	0.5454	0.54	0.8	0.59
14	0.6219	0.62	0.78	0.64
15	0.4928	0.59	0.72	0.56
16	0.6135	0.66	0.72	0.65
18	0.6304	0.73	0.73	0.66
19	0.5539	0.75	0.61	0.6

Updated Test-Retest Reliability:

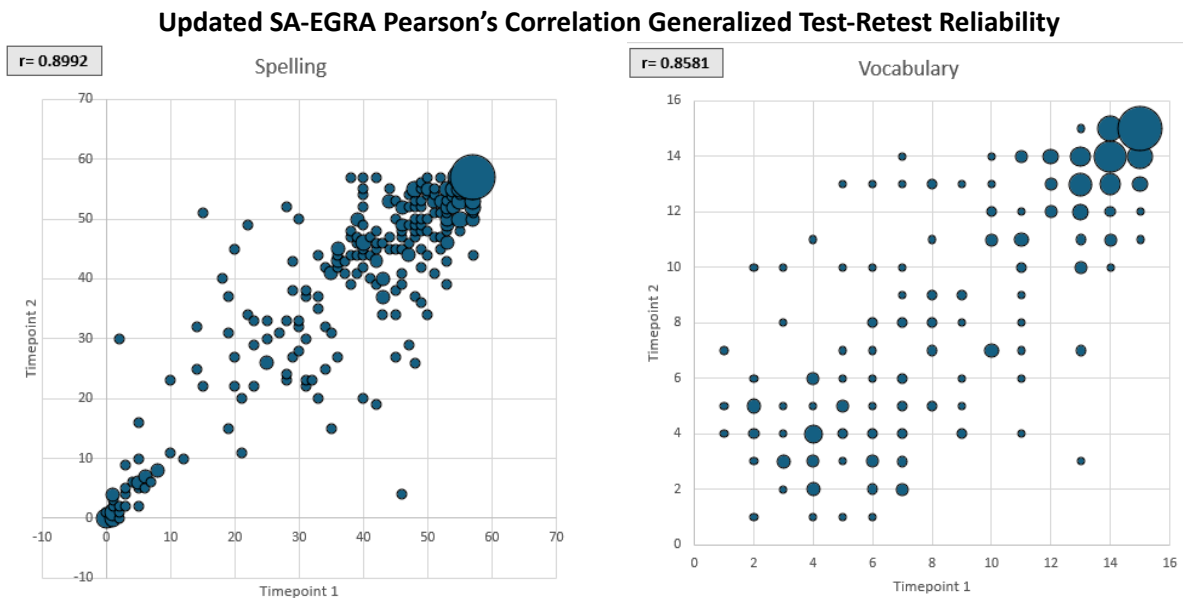
Updated Pearson's Correlation for the SA-EGRA Test-Retest

SA-EGRA Task Percent Score	Correlation
Letter Sounds	0.7519
Short Reading Comprehension	0.7297
Syllables	0.7314
Syntax	0.8529
Long Reading Comprehension	0.8229
Vocabulary	0.8581
Spelling	0.8992

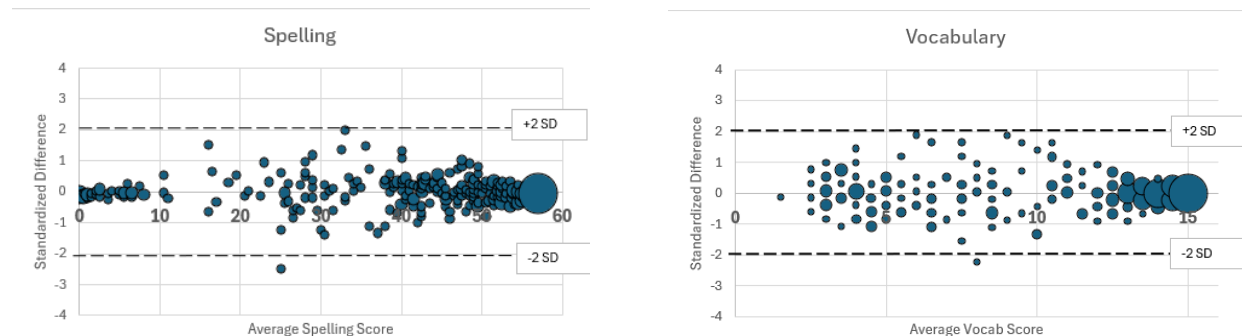
Test-retest reliability was assessed using Pearson's correlation to report the generalized relationship between student scores assessed across the two timepoints. The original analysis can be found in Exhibit 8 of the report on page 15. This updated analysis with the reduced set of

items for Vocabulary and Spelling still show strong, positive relationships, as the Pearson's coefficients are well above 0.5.

The Pearson's correlation for the Spelling and Vocabulary subtasks also show strong, positive relationships with coefficients well above the acceptable threshold of 0.5, seen in the figures below.



Updated SA-EGRA Bland-Altman Plots of Test-Retest Reliability



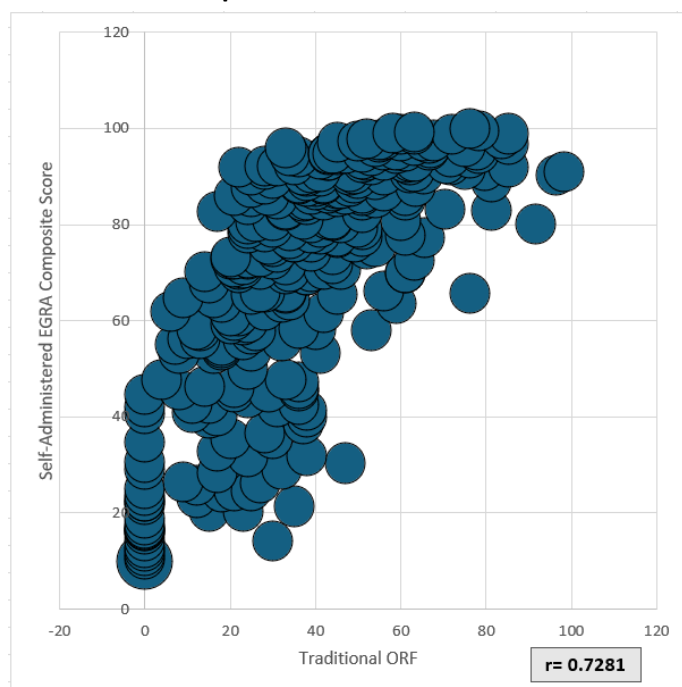
Bland-Altman plots were used to assess the level of agreement between pairs of repeated sets of results, as an additional measure of reliability for the test-retest analysis. In the initial analysis, the standardized difference between the students' scores at the two timepoints fell between ± 2 standard deviations in 95% of the data. This is still the case after the items were reduced for the subtasks, as can be seen here.

The original analysis of test-retest reliability for the assessment and for each subtask individually can be found in Annex D of the report.

Updated Construct Validity:

Lastly, construct validity was assessed to ensure the instrument is a valid measurement of literacy. This was assessed by comparing to students' scores on Oral Reading Fluency (ORF) using a traditional EGRA. The same composite score creation method was used with the updated subtask scores, and the results of the correlation analysis are shown below. There is very little difference in this outcome from the original. The coefficient still indicates a fairly strong positive linear association between the two tasks, but the predictive ability of the model is still lacking on the individual level, as previously reported. The original analysis can be found in Exhibit 12 on page 18 of the report.

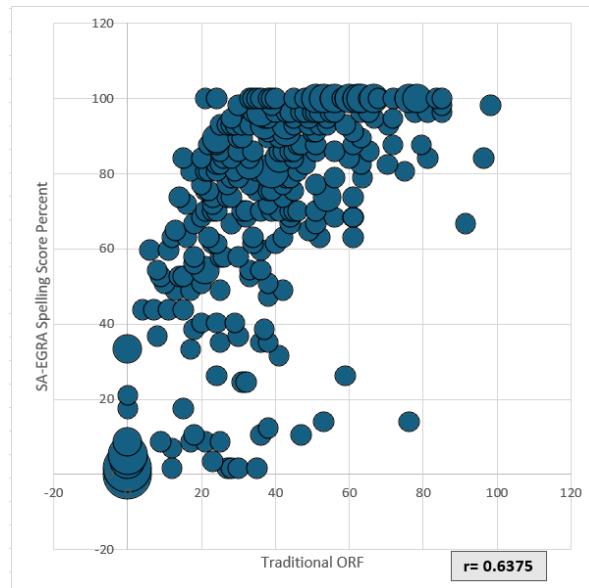
Updated SA-EGRA Composite Score vs. ORF on the Traditional EGRA



$$\text{SA_EGRA_composite} = 0.4 \times \text{spelling_total_score_pcnt} + 0.15 \times \text{short_read_comp_score_pcnt} + 0.1 \times \text{long_read_comp_score_pcnt} + 0.05 \times \text{letter_sounds_score_pcnt} + 0.1 \times \text{vocab_score_pcnt} + 0.1 \times \text{syntax_score_pcnt} + 0.1 \times \text{syllables_score_pcnt}$$

Additionally, Spelling was investigated to be used as a proxy measure for ORF, with analysis being reported in Annex E of the report. This analysis was repeated here, with similar outcomes. The positive linear association was slightly weaker than that of the composite score, but still had fairly strong statistical linking to the ORF on the traditional EGRA.

Updated SA-EGRA Spelling Score Percent score vs. Traditional EGRA Oral Reading Fluency



Conclusion:

This analysis was conducted to ensure that the removal of 4 items from the Spelling subtask and 5 items from the Vocabulary subtask would help to shorten the test without compromising reliability and validity. The above analysis shows that this is the case, and that while there are minor changes to the specific analyses, the conclusion of the initial Pilot analysis is still accurate. The SA-EGRA shows strong internal consistency as an assessment and within each of the tasks. Test-Retest analysis indicated that the SA-EGRA is a reliable measure of Kiswahili literacy for Grades 1-4 in Tanzania, and the SA-EGRA also shows strong construct validity when the composite score is compared to traditional ORF scores. With the minor changes, this Kiswahili SA-EGRA is still a reliable, valid instrument for assessing Early Grade Literacy in Grades 1-4 students in Tanzania.